

Time-Modulated Antenna Array Capable of Real-Time Tracking a Moving Target with Measurement Setup

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Abstract—This document summarizes the work of the WaveSpotters team for the 2025 IEEE AP-S Student Design Contest. We designed and implemented a Time Modulated Antenna Array (TMAA) as well as a complete measurement platform, both utilizing Zigbee-based devices. For antenna control and data acquisition a dedicated application was developed.

Index Terms—adaptive antennas, antennas, antenna arrays, time-modulated antenna arrays

I. INTRODUCTION

Given the task of developing an antenna system capable of real-time tracking of a moving target, we explored several beamforming architectures. However, due to strict budget constraints, realization of a conventional analog phased array or a digital beamforming system proved impractical. Therefore, we decided to base our design on an alternative method which utilizes periodic modulation of signals in the feed network using microwave switches. An antenna array in which this method is implemented is referred to as a time-modulated antenna array (TMAA) [1]–[4]. A distinctive feature of the TMAA is the use of time as an additional parameter in synthesizing the desired shape of the radiation pattern. The elements of the TMAA are being periodically connected and disconnected from the feeding network. As a result, the signal at the output of the feed network (in the case of a receiving TMAA) is a combination of signals received by different elements at different times. Due to the periodicity of the switching functions, the spectrum of a time-modulated signal is composed of a carrier component and harmonic components (sidebands). The timing of the on and off switching of the excitation signals has a critical impact on the shape of the radiation patterns. A Ka-band multi-beam phased-array transmitter is an excellent example of recent advancements in this field [5].

We have developed a sixteen-element TMAA operating in the 2.4 GHz ISM band, equipped with eight microwave switches that allow steering of the main beam in 4.5° steps in the azimuth plane, covering a total angular range of 118° .

II. THEORY OF OPERATION

The TMAA consists of $N = 8$ fundamental elements numbered $n = 0 \dots N - 1$. The fundamental element consists of two patch antennas with opposite orientations of the polarization vector, connected to a feed network via a single-pole double-throw (SPDT) microwave switch, as shown in Fig. 1. If a microwave signal of frequency f_c is incident perpendicularly upon an element, one of the antennas receives signal $s_n(t)$, while the second antenna receives the same signal but with an inverted phase $-s_n(t)$, due to the opposite arrangement of antennas.

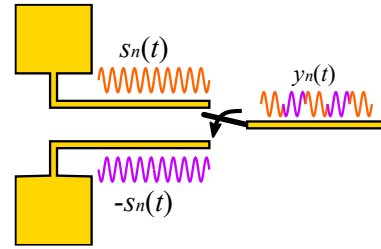


Fig. 1 Fundamental element of the TMAA

The switch operates periodically with period T_0 , so the output signal is equal to $+s_n(t)$ during the first half of the period and $-s_n(t)$ during the second half. The output signal from the switch can be described as a product of the input signal $s_n(t)$ and a rectangular bipolar function $m_n(t)$ with a high value of 1 and a low value of -1 .

$$y_n(t) = s_n(t)m_n(t) \quad (1)$$

Examples of signals at the output of the first four switches are shown in Fig. 2. Assuming that the modulation functions $m_n(t)$ are periodic, they can be represented as a combination of complex Fourier series coefficients $M_n^{(k)}$:

$$m_n(t) = \sum_{k=-\infty}^{\infty} M_n[k] e^{j2\pi f_0 k t} \quad (2)$$

$$M_n[k] = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} m_n(t) e^{-j2\pi f_0 k t} dt \quad (3)$$

where $M_n[k]$ is the k -th Fourier coefficient of the n -th modulating function with a period of T_0 and pulse repetition frequency $f_0 = 1/T_0$. Later, the index k , is referred to as the number of the spectral component. Fig. 3 shows example values of Fourier coefficients $M_n[k]$. The modulation process creates spectral replicas of the received signal at odd multiples of the modulation frequency.

Thanks to the time shift property of the Fourier transform, which states that a time delay in the time domain causes a phase shift in the frequency domain, the phase of the sidebands can be precisely controlled by delaying the modulating functions in the time domain:

$$m_n(t) = m_0(t - \Delta t_n) \quad (4)$$

$$M_n[k] = M_0[k] e^{-j2\pi k \Delta t_n / T_0} \quad (5)$$

In practice, this phase progression is realized by introducing time delays into the modulation waveforms. When no delay is applied (all switches are being toggled simultaneously), the beam is directed at 0° . By introducing a delay Δt between adjacent elements, a phase progression is imposed on the k -th sideband according to

$$\beta = 2\pi k \Delta t / T_0 \quad (7)$$

For example, $\Delta t = 5 \text{ ns}$ produces a 9° phase shift on the first sideband ($k = 1$). This enables beam steering similar to that of a conventional phased array, but without the need for microwave phase shifters, which are replaced by much simpler switches [1].

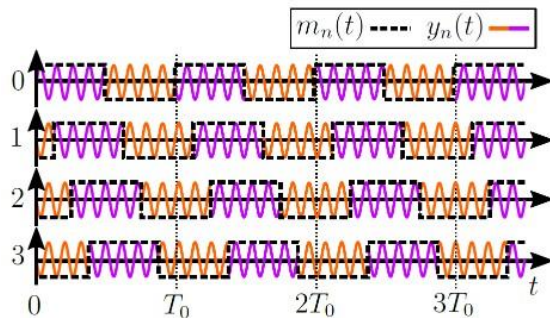


Fig. 2. Progressively delayed modulating functions and modulated signals

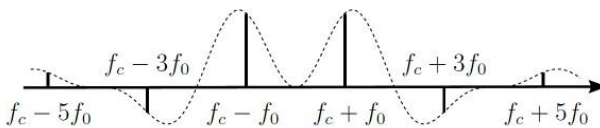


Fig. 3. Spectrum of bipolar modulating function $m_n(t)$

III. DEVELOPMENT OF TMAA SYSTEM

A. System Design

The developed TMAA's architecture is composed of two main functional modules, illustrated in Fig. 4: a control module and a radio-frequency (RF) module. Each module is realized on a 1-mm-thick FR-4 printed circuit board (PCB). Both modules are interconnected via plated through-holes. The first advantage is modular flexibility—each board can be swapped independently if a design error occurs. The second is lower production cost compared to a single four-layer PCB.

B. Control module

The function of the control module is to generate the drive signals for the RF switches. The modulation frequency was set to $f_0 = 5 \text{ MHz}$, placing the resulting sidebands at the center frequencies of adjacent channels in the ISM band (see Fig. 3). The 5 MHz, 50 %-duty rectangular waveform is generated by a miniature microelectromechanical system (MEMS) clock generator (Microchip DSC1001), selected due to its fast rise time ($t_r = 1 \text{ ns}$), fall time ($t_f = 0.9 \text{ ns}$), and high frequency stability ($\pm 50 \text{ ppm}$). The clock signal is fed into the first of eight cascaded programmable digital delay lines (Maxim Integrated DS1023S-500+). Each delay line can introduce a delay ranging from 0 ns to 1275 ns in 5 ns increments, yielding up to 255 distinct settings. Every delay line provides two outputs—a delayed output and an unaltered reference output. In the daisy-chain configuration, the reference output of one delay line serves as the input for the next, simplifying control-signal distribution. Since each RF switch requires complementary drive pulses (one logic high, one logic low), logic inverters generate the inverted version of each delay-line output.

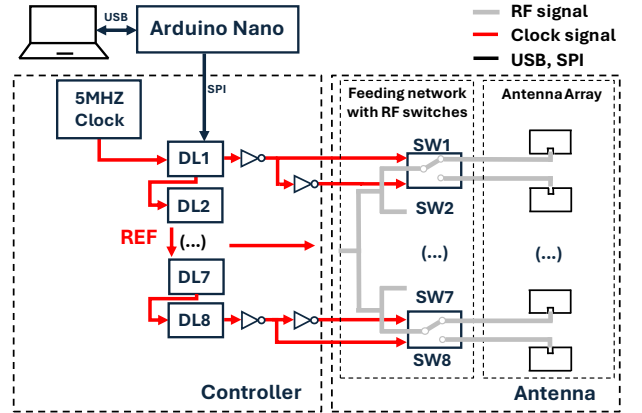


Fig. 4. Schematic of the TMAA

C. RF Module

The RF module of the TMAA comprises a sixteen-element patch-antenna array and a feeding network with switches. Each patch follows the dimensions shown in Fig. 5. Patch antennas are arranged in two parallel linear subarrays with a uniform inter-element spacing of 48 mm (half wavelength) as illustrated in Fig. 6. To toggle between the two antennas in each pair, Skyworks AS214 microwave switch was chosen for its 10%–90% rise/fall times of 10 ns—significantly shorter than the required switching interval $T_0 = 200 \text{ ns}$. At 2.45 GHz, the

manufacturer specifies an insertion loss of 0.4 dB and a port-to-port isolation of 25 dB. Each SPDT switch is driven by a complementary pair of control signals (one logic high, one logic low). The switching sequence for each element can be programmed independently. In one extreme, all switches toggle simultaneously, producing a broadside beam; alternatively, successive elements can be delayed by integer multiples of a defined Δt , thereby introducing a progressive phase shift across the array and enabling beam steering.

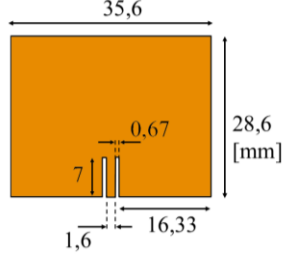


Fig. 5. Single patch antenna with its dimensions

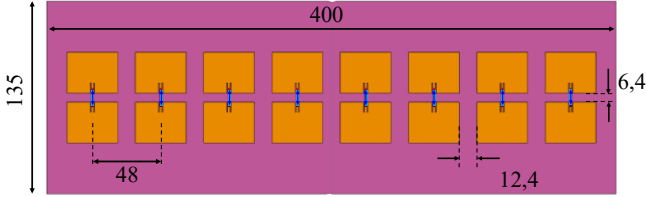


Fig. 6. Simulated model of the TMAA, dimensions shown in millimeters

D. Low-cost ISM transceivers

We employed off-the-shelf Digi XBee3 IEEE 802.15.4 (Zigbee) transceivers operating in the 2.4 GHz ISM band. Although these modules do not support a true continuous-wave direct-test mode, we achieve equivalent measurement fidelity by transmitting packets at a high repetition rate, which provides a quasi-continuous RSSI readout. Both the transmitter and receiver XBee3 modules are configured and controlled via Python scripts. The receiver module is directly interfaced with the TMAA for real-time RSSI sampling, while the battery-powered transmitter can be carried freely for tracking tests or fixed in place for pattern characterization, all without recourse to expensive lab instrumentation.

E. Web application

A custom web application provides full remote control over the TMAA system, allowing the user to switch seamlessly between beam-tracking and radiation-pattern measurement modes. All measurement data is automatically logged to a centralized database. Instructors or students can later query and visualize historical datasets via the same interface, enabling easy comparison of antenna behavior across different experiments without manual file handling or specialized software.

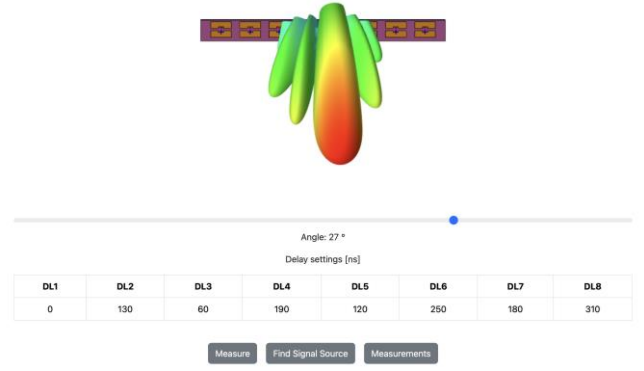


Fig. 7. Home page of the application showing current beam setting along with the values of delays across all delay lines

F. Electromagnetic Simulations

Full-wave electromagnetic simulations were conducted using the Method of Moments (MoM) in Altair FEKO. First, an isolated patch antenna was optimized to minimize its reflection coefficient and maximize gain. Once the single-patch performance had been validated, the array was modeled with each patch driven by its own independent excitation source. This allowed for simulations of beam steering by assigning each source a phase progression.

We simulated radiation patterns for phase progressions from -135° to $+135^\circ$ in 9° increments, corresponding to time-delay progressions from -75 ns to $+75$ ns in 5 ns steps. The results (Fig. 8) show that beam steering in 4.5° steps is achievable within a $\pm 59^\circ$ scan range, with directivity decreasing by 3 dB at the scan extremes. For beam steering angles of 0° , 30° , and 59° , the -3 dB beamwidths are 16.2° , 18.7° , and 25.5° , respectively.

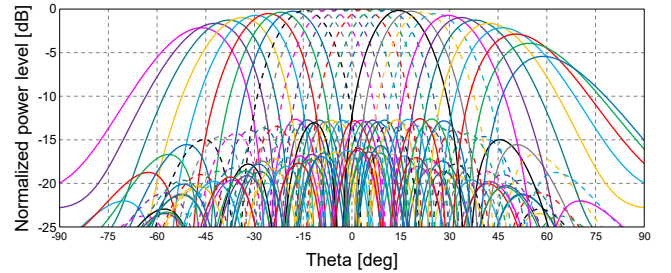


Fig. 8. Simulation results for all possible beam scanning settings

G. Measurements

Radiation pattern measurements of the TMAA array were carried out in an anechoic chamber using an SMBV100A signal generator, a Rohde & Schwarz FSP30 spectrum analyzer, and an RFspin RG18 reference antenna. The TMAA array was mounted on a turntable and measured in receive mode, while the signal generator and reference antenna served as the transmitting part of the measurement setup, used to emit a continuous-wave sinusoidal signal at 2.45 GHz. The TMAA was connected directly to the input of the spectrum analyzer. To monitor the variation in amplitude of the first harmonic (first sideband), the spectrum analyzer was configured with a center frequency of 2.455 GHz, zero span, and a sweep time of 20 seconds—corresponding to a 180-degree rotation of the azimuth positioner. Fig. shows the measured radiation patterns for selected beam directions (from -58.5° to 57.5° in 31.5°

steps). The measurement results closely match the simulated data, confirming the accuracy of the modeled beam-steering behavior.

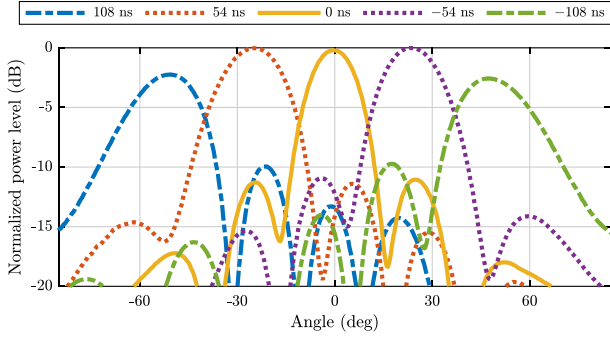


Fig. 9. Measured radiation patterns for selected beam directions

IV. BEAM-TRACKING CAPABILITIES

The tracking functionality is enabled by an adaptive population-based algorithm that, based on real-time RSSI measurements, dynamically adjusts the beam direction for best performance. The algorithm is based on evolutionary computation and operates with the following steps:

1. *Initialization.* An initial population of 3 beams directed toward -30° , 0° and 30° is defined.
2. *Fitness evaluation.* The beams are evaluated consecutively. Each beam is set on the TMAA and assigned a fitness value based on its RSSI value. This step repeats for each of the three beams defined in the initialization step.
3. *Selection.* The beam for which the highest RSSI is measured is considered to be closest to the direction of arrival. It is selected as the top individual Φ_{max} with its corresponding RSSI value of $RSSI_{max}$.
4. *Reproduction.* The top individual serves as the parent of a new population. It consists of a copy of the parent beam as well as two new beams with deviations determined by the RSSI value drop of the parent beam. If the RSSI drop is small, it indicates minimal movement of the signal source; thus, the new beams are generated with small deviations from the parent beam. If the RSSI drop is large, it suggests significant movement of the source, requiring the antenna to search a wider area, resulting in larger deviations for the new beams.

The new population obtained in the reproduction process is passed to step 2 (evaluation), where the beams are sequentially set on the TMAA. The algorithm operates in a loop and improves the RSSI value even under varying propagation conditions. These beams are defined by specific delay configurations for the SPDT switches, with each beam corresponding to one of the 31 pre-defined delay sets. The current beam shape is displayed on a monitor screen in the dedicated web application.

The beam-tracking performance of the developed TMAA was evaluated by determining both the maximum linear velocity at which reliable tracking could be maintained and the operational range over which the array could follow a moving transmitter. A signal source was moved along a circular path of

10 m radius, traversing from the first beam extremum (maximum scanning angle) to the second. Fig. 9 presents the resulting RSSI measurements for a walking speed of $0.3 \frac{m}{s}$: black color correspond to RSSI values recorded with the tracking algorithm disabled, while the green chart resembles RSSI recorded with the tracking algorithm enabled. When tracking was disabled, the received power exhibits clear fluctuations as the main beam and sidelobes of the TMAA sweep past the source. In contrast, with the tracking algorithm active, the RSSI remains approximately constant throughout the entire motion, demonstrating that the array continuously adjusts its main-lobe direction to maintain maximal received power.

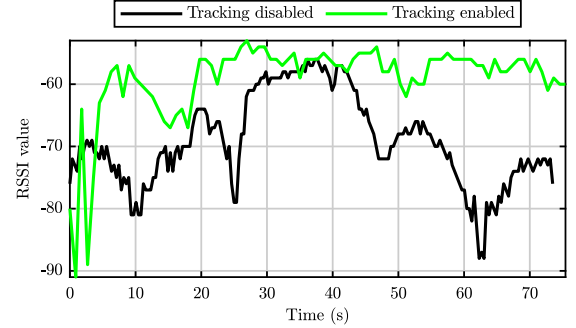


Fig. 9 Measurements obtained for a walking speed of 0.3 m/s

Fig. 10 shows the same measurements conducted under identical conditions, except with the transmitter moving at a constant walking speed of 1.2 m/s. After the algorithm “locks on” to the source, the RSSI values again remain stable.

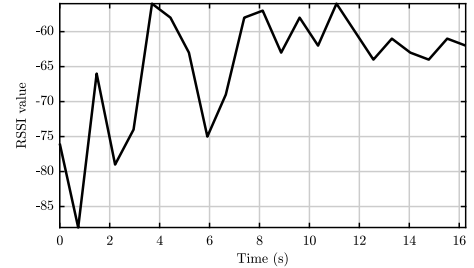


Fig. 10 Measurements obtained for a walking speed of 1.2 m/s

Furthermore, we tested the algorithm at a distance of 25 m and it continued to perform correctly; this experiment is documented in the accompanying video.

V. MEASUREMENT SYSTEM

In addition to real-time beam tracking, we implemented a low-cost radiation pattern measurement system using the same transmitter employed during tracking tests (Digi XBee3, IEEE 802.15.4). We can obtain a full H-plane radiation pattern—stepwise RSSI samples are collected as the antenna rotates on a motorized platform and streamed to the forementioned web application for beam-forming visualization. This approach removes the need for expensive laboratory instruments (spectrum analyzers, signal generators, VNAs). Fig. 11 shows a comparison between a radiation pattern measured with our system and one obtained with professional laboratory equipment.

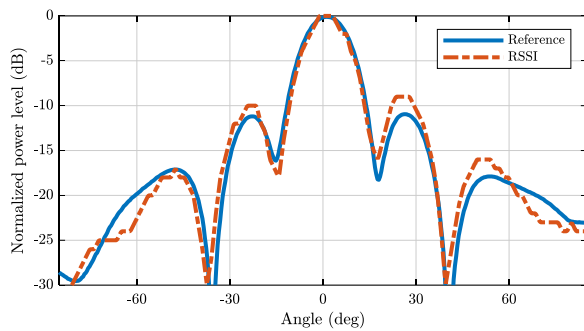


Fig. 11 Comparison of radiation patterns measured with professional instrumentation (reference) and self-developed measurement system (RSSI).

VI. USED PARTS AND MATERIALS

TABLE I

Product name	Qty.	Vendor	Cost (USD)
KYOCERA RF Capacitor 100pF	24	Mouser Electronics, Inc.	16.37
DSC1001DI1-005.0000	1	Mouser Electronics, Inc.	1.01
MEMS OSC LVCMOS 5MHz			
DS1023S-500+ Programmable Delay Line	8	Mouser Electronics, Inc.	184.15
AS214-92LF RF Switch	8	Mouser Electronics, Inc.	8.32
293D106X9010A2TE3 10 uF Tantalum Capacitor	8	Mouser Electronics, Inc.	5.69
C0603C104K8RACTU 0.1 uF Ceramic Capacitor	17	Mouser Electronics, Inc.	44.61
74HCT04D,653 CMOS Hex inverter	8	Mouser Electronics, Inc.	2.26
DRV8834 Motor Driver	1	Pololu	9.57
ABX00033 ARDUINO NANO	2	Mouser Electronics, Inc.	39.94
XK3-Z8S-WZM	1	DigiKey	98.21
TP4056 1-cel Li-Poly Battery Charger	1	TME	1.68
Rotary table	1	AliExpress	10.46
Total			422.27

The total price does not include PCB manufacturing costs, which amounted to 164 USD. Additionally, a 5% allowance has been included to cover spare components and 3D printing filament. As a result, the overall project cost is 607.10 USD, including 23% Polish value-added tax (VAT).

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Michał Bandyach (S'25) is a second-year Telecommunications student at Warsaw University of Technology. His work centers around RF and microwave circuit design, and high-frequency signal integrity. He specializes in PCB design. Vice president and co-founder of the Student Club of Antenna Design and a member of IEEE, AP-S, and ComSoc.



Radosław Rytel-Kuc is a second-year undergraduate student at the Warsaw University of Technology, majoring in Telecommunications. Passionate about programming, he is working with embedded systems, APIs, firmware, and hardware interfaces.



Wiktor Jastrzębski is a Telecommunications student at the Warsaw University of Technology, specializing in wireless technologies. His academic background encompasses RF circuit simulation, phased array systems, and impedance-matching techniques.



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Grzegorz Bogdan (S'16–M'20) was born in Poland in 1989. He received the B.E., M.E., and Ph.D. (with honors) degrees in Electrical and Computer Engineering with the major in Telecommunications from the Warsaw University of Technology (WUT), Warsaw, Poland, in 2012, 2013, and 2019, respectively. Since 2016 he has been with

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